

MethylDetector Theoretical Foundation

MethylPipeline

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1 Goal

`MethylDetector` identifies *differentially methylated positions* (DMPs) between two methylation centroids (e.g. healthy vs. disease) with:

- **Statistical rigour** — Histogram-derived Mann–Whitney testing with FDR correction.
- **Biological relevance** — A single canonical score (`effect_size`) that penalises positions where the distributions overlap heavily or where within-group variance is large.
- **Computational tractability** — A staged funnel that avoids expensive non-parametric testing and ECDF work on millions of positions that cannot pass the biological filter.

All distribution comparisons use the **ECDF only**; no Beta, Normal, or Beta-Binomial models are used.

2 Staged Pipeline

Stage 1 — Pre-filter (cheap). Start from positions present in both centroids after the coverage filter. Compute $|\Delta\mu| = |\hat{\mu}_1 - \hat{\mu}_2|$ from centroid means $\hat{\mu} = S_x/N$. Keep only positions with $|\Delta\mu| \geq \text{delta_mean_reduction}$.

Stage 2 — Centroid alignment handoff. Carry the shared genomic coordinates forward as per-centroid indices so every surviving position refers to the same locus in both centroids.

Stage 3 — Statistical significance test. For each aligned position, compute a p-value with Mann–Whitney from bin counts (Section 3.1).

Stage 4 — FDR correction. Two-stage Benjamini–Hochberg (`statsmodels fdr_tsbh`) on the pre-filtered position set.

Stage 5 — Statistical + heterogeneity filter. Retain positions with $q \leq \alpha$. Optionally drop positions where both groups exceed `max_tau2_for_dmp` (Section 4).

Stage 6 — Lazy ECDFView construction. Build `PchipInterpolator` objects only for the surviving DMP positions, using correct per-centroid indices.

Stage 7 — Continuous ECDF overlap and effect_size. Compute overlap (Section 5) and the canonical biological score (Section 6).

Stage 8 — Biological filter. Within each context, keep the minimum prefix whose cumulative `effect_size` reaches `effect_size_coverage` of the total effect mass.

Stage 9 — Optional biological-only rescue. Select non-significant but high-effect positions separately, flagged explicitly.

Stage 10 — Held-out top- k selection. Repeated stratified validation splits, cached ECDF log-likelihoods, and balanced accuracy choose the final top- k prefix.

3 Statistical Test

3.1 Mann–Whitney U from Histogram Bin Counts

The detector reconstructs the U statistic directly from the centroid `bin_counts` without raw sample values:

$$U = \sum_{b=1}^B \text{bc1}_b \cdot \text{CS2}_{b-1} + \frac{1}{2} \sum_{b=1}^B \text{bc1}_b \cdot \text{bc2}_b, \quad (1)$$

where bc1_b and bc2_b are the bin counts and $\text{CS2}_{b-1} = \sum_{k < b} \text{bc2}_k$ is the cumulative count of group 2 in all bins strictly below bin b . The first term counts concordant pairs, the second assigns half-credit to tied pairs.

Under the null hypothesis, U is approximately Normal with

$$\mathbb{E}[U] = \frac{N_1 N_2}{2}, \quad \text{Var}(U) = \frac{N_1 N_2}{12} \left[(N_1 + N_2 + 1) - \frac{\sum_b t_b(t_b^2 - 1)}{(N_1 + N_2)(N_1 + N_2 - 1)} \right], \quad (2)$$

where $t_b = \text{bc1}_b + \text{bc2}_b$ is the total count in bin b (tie correction). The z-statistic and two-sided p-value follow:

$$z = \frac{U - \mathbb{E}[U]}{\sqrt{\text{Var}(U)}}, \quad p = 2\Phi(-|z|), \quad (3)$$

where Φ is the standard Normal CDF.

3.2 Sample Variance for the Reliability Term

Sample variances are computed from stored sufficient statistics:

$$\hat{\sigma}^2 = \frac{S_{x^2} - S_x^2/N}{\max(N - 1, 1)}. \quad (4)$$

These variances are used in the biological reliability term and heterogeneity filters, not in the Mann–Whitney significance gate itself.

4 Between-Sample Heterogeneity (τ^2)

A DerSimonian–Laird-style decomposition uses the stored sufficient statistics (Σm , Σu , Σwx^2 , Σc^2 , N) to estimate between-sample variance without reading individual sample files.

Let $C = \Sigma m + \Sigma u$ (total coverage sum) and $\theta = \Sigma m/C$ (coverage-weighted pooled mean).

The within-group heterogeneity statistic is

$$Q = \Sigma wx^2 - \frac{(\Sigma m)^2}{C}, \quad (5)$$

and the between-sample variance estimate is

$$\hat{\tau}^2 = \max\left(0, \frac{Q - (N - 1)}{C - \Sigma c^2/C}\right). \quad (6)$$

Positions where both groups exceed `max_tau2_for_dmp` are dropped before the biological filter, because large $\hat{\tau}^2$ indicates heterogeneous sub-populations rather than a clean methylation shift.

5 ECDF Overlap

The continuous ECDF overlap between two class distributions is defined as

$$O = \int_0^1 \min(f_1(x), f_2(x)) dx, \quad (7)$$

where $f_1(x)$ and $f_2(x)$ are PCHIP-derived PDFs obtained by differentiating the spline CDF built from centroid `bin_counts`.

The integral is evaluated with the trapezoidal rule on a dense grid (default 512 points). PDFs are renormalised before integration to guard against small numerical drift in spline derivatives:

$$\tilde{f}_k(x) = \frac{f_k(x)}{\int_0^1 f_k(t) dt}, \quad k \in \{1, 2\}. \quad (8)$$

5.1 Discrete Bhattacharyya Approximation

In the fast path the overlap is approximated by the discrete Bhattacharyya coefficient computed directly from bin counts:

$$O_{\text{Bh}} = \sum_{b=1}^B \sqrt{p_{1b} \cdot p_{2b}}, \quad p_{kb} = \frac{\text{bc}_{kb}}{\sum_{b'} \text{bc}_{kb'}}. \quad (9)$$

This is $\mathcal{O}(n_{\text{pos}} \cdot B)$ and is used for the full position set before the statistical test reduces it.

6 Canonical Effect Size

The biological importance score is

$$\text{effect_size} = |\Delta\mu| \cdot (1 - O) \cdot \exp(-\lambda_{\text{var}}(\sqrt{\hat{\sigma}_1^2} + \sqrt{\hat{\sigma}_2^2})), \quad (10)$$

where:

- $|\Delta\mu| = |\hat{\mu}_1 - \hat{\mu}_2|$ — absolute mean methylation difference.
- O — continuous ECDF overlap from Equation (7); $1 - O$ is the distributional separation (0 = complete overlap, 1 = no overlap).
- $\exp(-\lambda_{\text{var}}(\sqrt{\hat{\sigma}_1^2} + \sqrt{\hat{\sigma}_2^2}))$ — reliability penalty that reduces the score for diffuse, heterogeneous positions.
- λ_{var} (default 2.0) — variance penalty strength.

Both variances are treated independently; no equal-variance assumption is made. The variances are the estimator from Equation (4) and act only through the biological reliability term, so a diffuse locus is down-weighted even though the significance gate itself remains non-parametric.

The score is clipped to $[0, 1]$ after computation.

6.1 Effect Size Coverage Biological Filter

Within each methylation context, positions are sorted by `effect_size` descending. The minimum prefix whose cumulative mass reaches the target fraction is retained:

$$k^* = \min \left\{ k : \frac{\sum_{i=1}^k e_i}{\sum_i e_i} \geq \text{effect_size_coverage} \right\}, \quad (11)$$

where e_i is the effect size of the i -th position after sorting.

7 ECDFClassifier

MethylDetector trains an **ECDFClassifier** on the selected biological DMPs.

7.1 Log-Likelihood

For a new sample with methylation fractions $x = (x_1, \dots, x_K)$ at the selected DMP positions, the log-likelihood for class k is

$$\log L(\text{class}_k \mid x) = \sum_{i=1}^K w_i \cdot \log F'_{k,i}(x_i), \quad (12)$$

where:

- $F'_{k,i}(x)$ is the PCHIP-derived PDF from the centroid of class k at DMP position i .
- $w_i = \text{effect_size}_i / \max_j(\text{effect_size}_j)$ — positions with larger effect contribute more.
- A small floor is applied: $F'_{k,i}(x) \geq 10^{-300}$ to avoid $\log(0)$.

7.2 Posterior and Prediction

The posterior probability of class k is obtained by temperature-scaled softmax:

$$P(\text{class}_k \mid x) \propto \exp\left(\frac{\log L(\text{class}_k \mid x)}{T}\right), \quad (13)$$

where $T \geq 0.1$ is the temperature parameter (default $T = 2.0$). Prediction is $\hat{k} = \arg \max_k P(\text{class}_k \mid x)$.

The ECDFClassifier stores the `bin_counts` histograms per DMP position (not α/β parameters), and uses the same PCHIP PDF model as the DMP detection stage, keeping detection and classification fully consistent.

8 FDR Correction

After computing per-position p-values, multiple-testing correction uses the two-stage Benjamini–Hochberg procedure (`statsmodels fdr.tsbh`).

Note on liberality. FDR is applied to the pre-filtered position subset (after `delta_mean_reduction`). Q-values are therefore liberal relative to testing all positions. For prostate-cancer-scale data (CG context: ≈ 2000 significant positions out of 4.3 million), the practical effect is small because the pre-filter removes positions that would not survive the biological filter regardless of significance.

9 Summary Table

Component	Role
Distribution model	ECDF only (<code>bin_counts</code> , $B = 20$ default)
Statistical test	Histogram-derived Mann–Whitney U
Multiple testing	Two-stage Benjamini–Hochberg (<code>fdr_tsbh</code>)
Heterogeneity filter	$\hat{\tau}^2$ from <code>Sc2</code> , <code>Swx2</code>
Pre-filter gate	<code>delta_mean_reduction</code> (before the test)
Overlap	Continuous ECDF: $\int_0^1 \min(f_1, f_2)$
Biological score	Canonical <code>effect_size</code> , Eq. (10)
Biological filter	Per-context effect-size coverage mass selection
Rescue track	Optional non-significant high-effect loci
Validation	Stratified holdout BA with ECDF log-likelihood prefixes
Classifier	ECDFClassifier: PCHIP PDF log-likelihood, weighted

Table 1: Summary of the MethylDetector pipeline components.